

Customer Retention Analysis Using Engagement and Product Utilization

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CHAPTER 1

ABSTRACT

Customer retention has become one of the most important strategic objectives in the banking industry. The increasing competition among financial institutions has made it difficult to retain customers for longer periods. Traditional churn analysis mainly focuses on demographic and financial indicators such as customer age, income, and account balances. However, modern banking research suggests that customer engagement and product utilization play a much more significant role in determining long-term customer loyalty.

This study investigates the relationship between customer engagement, product utilization, and customer retention using the European Bank Customer Dataset containing 10,000 customer records. The research examines customer activity levels, product adoption patterns, account balances, and behavioral characteristics to identify factors that influence customer churn.

The analysis reveals that inactive customers are significantly more likely to churn than active customers. Customers with only one banking product also demonstrate higher churn rates compared to customers with two products. Furthermore, the study identifies a large group of high-balance but disengaged customers who represent significant financial risk to the bank.

The findings of this research provide valuable insights for designing engagement-driven retention strategies, personalized product recommendations, customer loyalty programs, and cross-selling initiatives. The study demonstrates that behavioral analytics can substantially improve customer retention and support data-driven decision-making in the banking sector.

CHAPTER 2

INTRODUCTION

The banking industry has undergone significant transformation due to digitalization, increasing competition, and changing customer expectations. Banks are no longer competing only on financial products and interest rates; they are increasingly competing on customer experience and relationship quality. As a result, customer retention has become a strategic priority for financial institutions worldwide.

Customer churn refers to the process by which customers discontinue their relationship with a bank. High customer churn negatively affects several important business metrics including customer lifetime value, revenue stability, profitability, and long-term competitiveness. Acquiring new customers is often significantly more expensive than retaining existing customers. Therefore, understanding the factors that influence customer retention is essential for sustainable business growth.

Traditionally, banks have relied heavily on demographic information such as age, salary, and credit scores to predict customer behavior. However, recent studies indicate that customer behavior and engagement patterns provide much stronger indicators of long-term loyalty. Customers with high balances or high salaries may still choose to leave the bank if they are not actively engaged with banking services or if they use only a limited number of products.

Customer engagement can be defined as the degree of interaction and relationship between a customer and the bank. Product utilization refers to the number and variety of banking products used by customers. Together, these factors provide valuable insights into relationship strength and retention potential.

This research focuses on analyzing customer engagement and product utilization to identify customer segments, evaluate retention patterns, and provide actionable recommendations for improving customer loyalty and reducing churn.

2. BACKGROUND OF THE STUDY

Banks increasingly recognize that customer behavior and engagement are stronger indicators of retention than demographic characteristics alone. Customers may appear financially valuable due to high balances or high salaries but may still exhibit a high probability of leaving the bank because of low engagement levels and limited product adoption.

Understanding customer engagement patterns is essential for several business activities including:

- Customer retention initiatives.
- Cross-selling and up-selling strategies.
- Customer loyalty programs.
- Personalized banking services.
- Relationship management.

The modern banking environment requires institutions to move beyond traditional financial analysis and adopt behavioral analytics approaches that focus on understanding customer needs, preferences, and interactions.

This study investigates customer engagement and product utilization from a behavioral perspective and aims to provide a comprehensive understanding of customer retention drivers.

CHAPTER 3

PROBLEM STATEMENT

Despite possessing large amounts of customer data, banks often struggle to identify the behavioral factors that influence customer retention. Most retention strategies remain generic and fail to address customer-specific needs and engagement patterns.

Banks frequently lack:

- Quantitative insights regarding customer engagement and retention.
- Clear understanding of how product utilization influences loyalty.
- Methods for identifying high-value but disengaged customers.
- Evidence regarding whether high account balances guarantee customer retention.

As a result, retention initiatives become reactive rather than proactive, increasing customer acquisition costs and reducing overall profitability.

This research attempts to bridge this gap by analyzing customer engagement and product utilization patterns and their relationship with customer churn.

CHAPTER 4

PROJECT OBJECTIVES

The primary objective of this research is to analyze customer engagement and product utilization patterns and evaluate their impact on customer retention and loyalty in the banking industry.

4.1 Primary Objectives

- To evaluate the relationship between customer engagement and customer churn.
- To measure the impact of product count and product utilization on customer retention.
- To identify high-value customers who are financially valuable but exhibit low engagement levels.

4.2 Secondary Objectives

- To support engagement-driven retention strategies.
- To improve product bundling and cross-selling decisions.
- To identify silent churn among premium customers.
- To provide actionable recommendations for customer relationship management.
- To improve customer lifetime value through personalized engagement initiatives.

This study aims to help financial institutions make data-driven decisions regarding customer retention and relationship management.

CHAPTER 5

DATASET DESCRIPTION

The study uses the European Bank Customer Dataset containing customer demographic, financial, and behavioral information. The dataset consists of 10,000 customer records and 14 variables.

Column Name	Description
CustomerId	Unique customer identifier
Surname	Customer surname
Credit Score	Customer creditworthiness score
Geography	Customer region (France, Germany, Spain)
Gender	Male / Female
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
No of Products	Number of banking products used
HasCrCard	Credit card ownership status
Is Active Member	Customer activity status
Estimated Salary	Estimated annual salary
Exited	Customer churn indicator (1 = Churned, 0 = Retained)

The dataset consists of 10,000 customer records and includes both categorical and numerical variables. These attributes were used to evaluate engagement levels, product utilization behavior, premium customer risk, and retention performance across different customer segments.

CHAPTER 6

RESEARCH METHODOLOGY

The research methodology adopted in this study consists of multiple stages including data preparation, exploratory data analysis, customer segmentation, KPI analysis, and dashboard development.

6.1 Data Collection

The European Bank Customer Dataset was collected and imported into Python for analysis.

6.2 Data Validation

The following validation activities were performed:

- Missing value identification.
- Duplicate record detection.
- Data type verification.
- Validation of binary variables.

6.3 Data Cleaning

The following preprocessing activities were performed:

- Removal of non-informative columns.
- Validation of customer engagement fields.
- Validation of churn labels.

6.4 Exploratory Data Analysis

The following analyses were conducted:

- Customer churn distribution.
- Customer engagement analysis.
- Product utilization analysis.
- Financial commitment analysis.
- Customer segmentation.

6.5 KPI Development

The following KPIs were developed:

- Engagement Retention Ratio.
- Product Depth Index.
- High-Balance Disengagement Rate.
- Credit Card Stickiness Score.
- Relationship Strength Index.

6.6 Dashboard Development

Interactive dashboards were developed using Tableau/Power BI to visualize customer engagement patterns and retention drivers.

CHAPTER 7

DATA VALIDATION AND PREPROCESSING

Before performing the analysis, the dataset was validated and cleaned.

7.1 Missing Values Analysis

The analysis revealed that the dataset contained no missing values.

Table 1: Missing Values Summary

Variable	Missing Values
All Variables	0

7.2 Duplicate Records

The dataset contained no duplicate records.

Table : Duplicate Analysis

Metric	Duplicate Rows
Value	0

7.3 Data Type Validation

The dataset contained:

- Numerical variables.
- Categorical variables.
- Binary variables.

All variables were found to be consistent and suitable for analysis.

7.4 Feature Engineering

The following analytical variables were created:

- Customer Profile.
- Product Category.
- Relationship Strength Index.
- Customer Segmentation Variables.

The data validation process confirmed that the dataset was reliable and ready for exploratory analysis.

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EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was conducted to understand customer behavior, engagement patterns, and retention drivers.

8.1 Customer Churn Distribution

Out of 10,000 customers:

- Retained Customers: 7,963 (79.63%)
- Churned Customers: 2,037 (20.37%)

This indicates that approximately one-fifth of customers have left the bank, highlighting the importance of retention strategies.

8.2 Customer Engagement Analysis

Inactive customers exhibited significantly higher churn rates compared to active customers.

- Inactive Customer Churn Rate:
26.85%
- Active Customer Churn Rate:
14.27%

The findings indicate that customer engagement plays a critical role in customer retention.

8.3 Product Utilization Analysis

Customers using two products exhibited the highest retention rates, while customers with three or more products demonstrated extremely high churn rates.

This suggests that moderate product adoption strengthens customer relationships, while excessive product complexity may negatively impact customer satisfaction.

8.4 Financial Commitment Analysis

The analysis identified:

- 1,247 high-balance but disengaged customers.
- 2,484 salary-balance mismatch customers.

These customer segments represent significant opportunities for personalized retention initiatives and cross-selling campaigns.

CHAPTER 9

ANALYTICAL METHODOLOGY

The analytical methodology for this project was designed to evaluate customer engagement, product utilization, and retention behavior using banking customer data. Multiple analytical approaches were applied to identify churn patterns, high-risk customer segments, and relationship-strength indicators.

9.1 Engagement Classification

Customers were categorized based on their engagement level and banking activity status. The following customer groups were identified:

- Active engaged customers
- Inactive disengaged customers
- Active customers with low product usage
- High-balance inactive customers

This classification helped in understanding how customer activity influences retention behavior and churn risk.



Figure 9.1: Customer Engagement Classification and Distribution

9.2 Product Utilization Analysis

Product usage patterns were analyzed to evaluate their impact on customer loyalty and retention.

The following analyses were performed:

- Churn rate by number of products
- Single-product vs multi-product customer retention
- Relationship between product depth and churn probability

This analysis helped determine whether increased product adoption strengthens customer relationships with the bank.

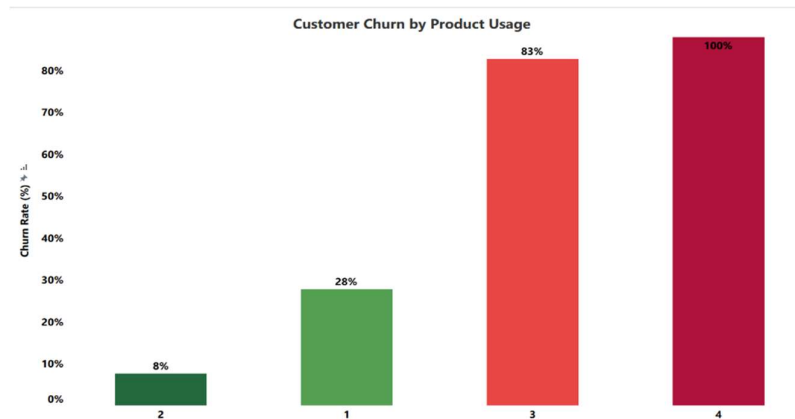


Figure 9.2: Impact of Product Utilization on Customer Churn

9.3 Financial Commitment vs Engagement Analysis

Customer financial behavior was analyzed using balance and salary-related attributes.

The following assessments were conducted:

- Balance vs customer activity analysis
- Salary and balance comparison
- Identification of high-value disengaged customers
- Detection of premium customers at risk of churn

This helped identify financially strong customers with weak engagement levels.

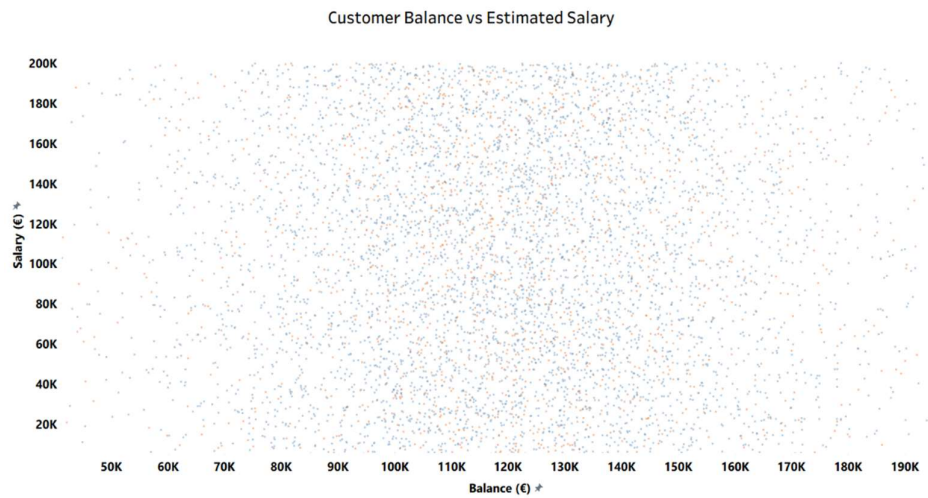


Figure 9.3: Financial Commitment and Customer Engagement Analysis

9.4 Retention Strength Assessment

Retention strength indicators were evaluated to identify stable and loyal customer segments.

The following factors were considered:

- Customer engagement level
- Product utilization depth
- Customer activity status
- Relationship stability across customer groups

This assessment helped define “sticky customer” profiles and identify engagement behaviors associated with long-term customer retention.



Figure 9.4: Distribution of High-Risk Premium Customers

CHAPTER 10

KEY PERFORMANCE INDICATORS (KPIs)

Several Key Performance Indicators (KPIs) were developed to measure customer engagement, relationship strength, and retention behavior.

10.1 Engagement Retention Ratio

The Engagement Retention Ratio measures the difference in churn rates between active and inactive customers.

Results:

- Active Customer Churn Rate: 14.27%
- Inactive Customer Churn Rate: 26.85%

The findings indicate that inactive customers are nearly twice as likely to leave the bank compared to active customers.

10.2 Product Depth Index

The Product Depth Index evaluates customer loyalty based on the number of products used.

Results:

- One Product: 27.71% churn
- Two Products: 7.58% churn
- Three Products: 82.71% churn
- Four Products: 100% churn

The analysis suggests that customers using two products demonstrate the strongest relationship with the bank.

10.3 High-Balance Disengagement Rate

The analysis identified 1,247 high-balance but disengaged customers. These customers represent a significant financial risk because they possess high balances but exhibit low engagement.

10.4 Credit Card Stickiness Score

The analysis revealed that credit card ownership has only a minor impact on customer retention and does not significantly reduce churn probability.

10.5 Relationship Strength Index

A Relationship Strength Index was developed using customer activity status, product count, and account balance.

The majority of customers demonstrated a medium relationship strength score, indicating opportunities to improve engagement and deepen customer relationships.

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CUSTOMER SEGMENTATION ANALYSIS

Customer segmentation was performed to identify customer groups with different engagement characteristics and retention risks.

The following customer segments were identified:

Table 4: Customer Segmentation

Customer Segment	Number of Customers
Active Engaged Customers	2588
Active Low Product Customers	2563
Inactive High Balance Customers	2456
Inactive Disengaged Customers	878
Other Customers	1515

The segmentation analysis revealed that inactive high-balance customers represent one of the most important customer groups because they possess significant financial value while simultaneously demonstrating low engagement levels.

Similarly, active low-product customers represent strong cross-selling opportunities because they are engaged but have limited product adoption.

The segmentation framework provides valuable support for personalized retention campaigns and relationship management strategies.

CHAPTER 12

FINDINGS AND DISCUSSION

The study generated several important findings regarding customer retention and engagement.

Key Findings:

1. Approximately 20.37% of customers have churned.
2. Inactive customers exhibit significantly higher churn rates than active customers.
3. Customer engagement has a stronger influence on retention than financial strength alone.
4. Customers using two products demonstrate the highest loyalty and lowest churn rates.
5. Customers using three or four products exhibit extremely high churn rates.
6. A large number of high-balance but disengaged customers were identified.
7. More than 2,400 customers exhibit salary-balance mismatches and represent potential growth opportunities.
8. Credit card ownership does not significantly improve customer retention.
9. Customer behavior and engagement patterns provide stronger retention indicators than demographic variables.

These findings confirm that customer engagement and product utilization should be central components of retention strategies in modern banking institutions.

CHAPTER 13

RECOMMENDATIONS

Based on the findings of this study, the following recommendations are proposed:

1. Increase customer engagement initiatives through personalized communication and digital interaction strategies.
2. Develop targeted retention campaigns for inactive customers.
3. Focus cross-selling initiatives on customers using only one banking product.
4. Investigate the causes of extremely high churn among customers with three or more products.
5. Implement personalized engagement programs for high-balance inactive customers.
6. Improve customer relationship management through behavioral analytics.
7. Develop loyalty programs focused on increasing customer engagement and product adoption.
8. Continuously monitor customer engagement indicators and retention KPIs.

The implementation of these recommendations can significantly improve customer retention and strengthen long-term customer relationships.

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BUSINESS IMPACT

The findings of this project provide significant business value by helping banks better understand customer behavior, engagement patterns, and retention risk factors. The analysis supports data-driven decision-making and enables financial institutions to design more effective customer retention strategies.

❖ Key Business Impacts

1. Improved Customer Retention

The identification of engagement-related churn factors helps banks implement targeted retention strategies for high-risk customers.

2. Better Customer Relationship Management

Analyzing customer engagement and product utilization improves relationship-strength assessment and customer segmentation.

3. Increased Product Adoption

The study highlights the importance of multi-product usage, supporting cross-selling and product bundling opportunities.

4. Reduction in Premium Customer Churn

The identification of high-balance inactive customers enables banks to protect financially valuable customers through personalized retention initiatives.

5. Enhanced Decision-Making

Interactive dashboard analytics provide management with clear insights for strategic planning and customer retention optimization.

6. Stronger Engagement-Driven Strategy

The project demonstrates that behavioral engagement is a stronger retention indicator than financial status alone, encouraging banks to focus on customer interaction and loyalty-building programs.

Overall, this analysis helps banks improve customer loyalty, reduce churn risk, optimize engagement strategies, and strengthen long-term business performance.

CHAPTER 15

CONCLUSION

This study investigated customer engagement and product utilization from a behavioral and relationship-strength perspective. The results demonstrate that customer engagement plays a critical role in customer retention and significantly influences customer churn.

The analysis revealed that inactive customers are substantially more likely to leave the bank compared to active customers. Product utilization also exhibited a strong relationship with customer retention, with customers using two products demonstrating the highest levels of loyalty.

Furthermore, the identification of high-balance but disengaged customers highlights the importance of behavioral analytics in customer relationship management.

The findings suggest that financial strength alone does not guarantee customer loyalty. Instead, customer engagement, relationship depth, and product utilization provide stronger indicators of long-term retention.

This research provides valuable insights for banks seeking to design engagement-driven retention strategies, improve customer experience, and strengthen long-term competitiveness in the banking industry.

CHAPTER 16

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